

The adoption of cloud computing in small and medium enterprises: a developing country perspective

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Abstract

Purpose – The purpose of this paper is to examine the determinants of cloud computing adoption in small and medium enterprises (SMEs), further, to measure the effect of cloud computing adoption on cloud-supported firm performance through enhancing organisational agility.

Design/methodology/approach – The research model is developed by combining two popular theoretical models, namely, the unified theory of acceptance and use of technology (UTAUT) and the technology–organisation–environment (TOE) framework. Data are collected from 372 SMEs to test the model. The strengths of widely used structural equation modelling (SEM) are applied to analyse the data.

Findings – This study reveals that the significant predictors of cloud computing adoption are performance expectancy; effort expectancy; absorptive capacity; data security and privacy; and perceived trust. Also, cloud computing adoption positively influences firm performance directly and through organisational agility. The results of importance–performance map analysis indicate that effort expectancy falls in the critical zone, which needs to be improved.

Originality/value – This research is one of few that blends the strengths of UTAUT and TOE framework. The research outcomes yield noteworthy suggestions to cloud providers, managers and government policymakers on ways of motivating the spread of cloud computing in developing countries.

Keywords UTAUT, SMEs, TOE, Firm performance, Cloud computing adoption, Organizational agility

Paper type Research paper



1. Introduction

Cloud computing is an innovative computing model that uses the concepts of distributed and parallel computing to provide prevalent, convenient and on-demand access to the virtualised and distributed various computing resources such as hardware, software, networks, servers, storage and databases (Ebadi and Jafari Navimipour, 2019). It provides organisations with a way to enhance the capacity or add capabilities without investing in new infrastructure, training new personnel or licensing new software as it offers rented services, which leads to adjusting the level of usage according to the current needs (Subashini and Kavitha, 2011). As SMEs lack adequate technical capabilities to achieve their

expected outcomes, cloud services can ensure a level-playing field for them by creating opportunities to use state-of-the-art technologies that previously were unaffordable (Marston *et al.*, 2011). The unlimited flexibility, better reliability and high security offered by cloud computing enable SMEs to customise services and data for higher availability (Doherty *et al.*, 2015). Further, cloud computing adds values in both strategic and operational levels such as increasing annual revenue and minimising operational cost (Carcary *et al.*, 2014). Thus, the use of cloud computing can improve business operations and has provided genuine benefits for SMEs, which allows them to be competitive in the current dynamic business environment (Carcary *et al.*, 2014).

Despite the benefits of cloud computing, the rate of acceptance and use of this technology is still unsatisfactory, particularly among SMEs (Kumar *et al.*, 2017). The major challenges towards the adoption of cloud technologies include lack of industry-specific standards; the paucity of technology readiness (TR); unawareness about potential benefits of this technology; lack of security and privacy; and the lack of trust (Lian *et al.*, 2014). The modest investment in technology by SMEs may affect their decision for not adopting cloud computing (Abdollahzadehgan, 2013). In addition, potential risks related to the use of cloud services affect cloud computing adoption. SMEs always remain to worry about the privacy and security of their data stored in the third-party's server. These concerns inhibit them from accepting and using cloud computing.

Although cloud computing is getting popular to the researchers, understanding the adoption behaviour of SMEs of cloud computing has not been sufficiently addressed in the literature (Sharma *et al.*, 2016), particularly in the developing country context. We found few studies in the literature which can be further classified into three categories. First, we found some technical studies which primarily focused on concepts and architectures of cloud computing (Chen and Deng, 2009; Jadeja and Modi, 2012; Youseff *et al.*, 2008). Second, some descriptive studies were found which generally stated the definition, types, benefits, risks and challenges of cloud computing (Carroll *et al.*, 2011; Catteddu, 2009). Third, few theoretically grounded studies were identified that investigate the determinants of cloud computing adoption in SMEs (Gupta *et al.*, 2013; Alshamaila *et al.*, 2013; Doherty *et al.*, 2015; Pathan *et al.*, 2017; Ooi *et al.*, 2018). These studies were criticised for applying exploratory research (Doherty *et al.*, 2015), using small sample size (Alshamaila *et al.*, 2013) and focusing developed countries (Agostino *et al.*, 2013). Moreover, most of the studies primarily have focused the initial adoption of cloud computing in SMEs; there are few empirical studies on the utilisation of cloud services in the literature. Specifically, the effect of cloud service adoption on firm performances (FPs) for cloud-supported operations and processes is not sufficiently addressed in the prior studies. Therefore, this research aims to examine the predictors of cloud computing adoption in SMEs, and further to assess the impact of cloud computing adoption on cloud-supported competitive performance.

This paper has developed the research model by combining two popular technology adoption models, namely, unified theory of acceptance and use of technology (UTAUT) (Venkatesh *et al.*, 2003) and technology–organisation–environment (TOE) framework (Tornatzky *et al.*, 1990). Further, we extend our model to assess the impact of cloud adoption on competitive FP. Many prior empirical studies claim that information technology (IT) assimilation does not affect the FP directly, rather their relationship is mediated by other higher order capabilities (Liu *et al.*, 2013). For example, Felipe *et al.* (2019) claimed that information systems (IS) capabilities influence FP through organisational agility (ORGA). Similarly, Sambamurthy *et al.* (2003) revealed that knowledge management and agility are two mediators of IT capabilities and FP relationship. Thus, we extend our model by adding ORGA as a mediator of cloud adoption and cloud-affected FP. Structural equation modelling

(SEM) has been applied to test the model. SEM is a popular and widely accepted statistical method for testing and estimating causal relationships using a mix of statistical data and qualitative causal assumptions (Nascimento *et al.*, 2018). Additionally, this study performed importance–performance map analysis (IPMA) in SEM to identify the critical factors that need special managerial attention to improve cloud computing adoption and FP.

The article proceeds with the explanation of the theoretical background. Next, based on the extensive review of the literature, this study develops the research model and hypotheses. After that, research methodology is depicted. Then, results, discussion and implications of the study are presented. Finally, we present the limitations and future research directions in this area.

2. Background

2.1 Cloud computing

Cloud computing evolved through the significant developments in hardware, distributed computing, and service delivery over network (Oliveira *et al.*, 2014). Armbrust *et al.* (2010) defined cloud computing as a network-based technology that delivers on-demand access to users for remotely available IT-enabled resources and capabilities. In other words, it refers to various computing application services such as email, office software and enterprise resource planning and ubiquitous resources that can be shared by the business employee or trading partners (Low *et al.*, 2011). Cloud-based solutions transform existing business processes, lower IT costs, provide real-time applications and offer access to ubiquitous storage (Luo *et al.*, 2011; Sultan, 2010; Armbrust *et al.*, 2010). This technology offers organisations and users convenient access to powerful computing through the network at minimal cost (Marston *et al.*, 2011). In general, cloud-based services include Software as a Service (SaaS) (e.g. Joyent and Salesforce CRM), Platform as a Service (PaaS) (e.g. Salesforce, Google AppEngine and Microsoft Azure) and Infrastructure as a Service (IaaS) (e.g. Amazon Elastic Compute Cloud (EC2), Rackspace, Amazon Simple Storage Service (S3) and GoGrid) on a pay-for-use basis (Park and Ryoo, 2013). Cloud computing can help converging IT efficiency and business agility of the firms (Oliveira *et al.*, 2014). The ways of achieving IT efficiency include using scalable hardware and software resources (Marston *et al.*, 2011), improving work efficiency (Low *et al.*, 2011) and ensuring highly available services (Armbrust *et al.*, 2010). The business agility of cloud computing includes the capacity of deploying mass computing technology quickly, minimising capital cost directly and responding rapidly to the market change (Oliveira *et al.*, 2014).

2.2 Organisational agility

ORGA is generally defined as a firms' unique capability which helps to identify and address uncertain events and changes in the business arena by reacting promptly and exploiting changes as further opportunities for expansion (Van Oosterhout *et al.*, 2006). Three interrelated dimensions of ORGA (e.g. customer agility, partnering agility and operational agility) are profoundly presented in the literature (Sambamurthy *et al.*, 2003).

Operational agility refers to the extent to which firms can easily and quickly retool their business operations to adapt to the market environment (Tallon, 2008). It highlights the need for a firm to detect changes, opportunities and threats in the environment and to provide swift and focused responses to customers and stakeholders by reconfiguring internal resources and processes (Mathiyakalan *et al.*, 2005). Customer agility allows firms to capitalise on changes in their market by delivering new or enhanced value propositions and helps improve internal operation adjustments to ensure customer retention (Lu and Ramamurthy, 2011). By improving their customer agility, firms manage to expand their

range of competitive actions and are therefore more prone to flourish in market positioning. Partnering agility is defined as the ability of an organisation to leverage the assets, knowledge and competencies of suppliers, distributors, contract manufacturers and logistics providers through alliances, partnerships and joint ventures (Felipe *et al.*, 2019). Partnering agility enables an organisation to modify or adapt its extended enterprise network when it needs access to assets, competencies or knowledge not currently available in its networks (Sambamurthy *et al.*, 2003).

These three dimensions are reinforcing each other as they are interrelated, and agile firms are smart enough to execute higher capability from them (Ravichandran, 2018). Current business environments are unpredictable in terms of their dynamism, turbulence and difficulty. The ability of ORGA helps them to chalk out the need for growth from outside and inside, develop an action plan for implementing new changes and achieve the desired goal (Worley and Lawler, 2010). The proper deployment of cloud technology can provide organisations with prompt access to IT resources and ensure a fast and smooth way of service, which enhances the ORGA of the firm (Sultan, 2011).

2.3 Firm performance

The fundamental objective of managing a firm is to achieve superior performance compared to its competitors. Undoubtedly, performance measurement is considered as one of the popular and recurrent topics in the literature of management (Felipe *et al.*, 2019). Nevertheless, there is a controversy regarding the definition, dimension and measurement of performance (Santos and Brito, 2012). In the literature, scholars have measured the return on investment of the firms from two perspectives: financial and non-financial (Chan and Chong, 2012; Ilmudeen and Bao, 2018). Financial performance not only relies wholly on the firm's efficiency itself but also on the market where the firm operates (Felipe *et al.*, 2019). For this reason, not all the financial ratios are equally accurate and meaningful while assessing the financial performance of an organisation. FP can be examined by the following two approaches: objective and subjective. According to objective approach, performance is measured through the indicators of financial performance such as revenue, profit margin, sales growth, return on asset (ROA), return on equity (ROE), return on investment (ROI), capital adequacy, liquidity ratio and stock prices. On the other hand, in the subjective approach, non-financial measures of operational performance are applied to judge the FP. Subjective indicators are developed with the perceptions of the respondents concerning how well their firm is performing, such as market share, product or service quality, corporate growth, novel or innovative products or services introduction (Felipe *et al.*, 2019; Singh *et al.*, 2016). The addition of subjective non-financial measures offers a wider perspective of operational factors that are placed behind the financial performance (Venkatraman and Ramanujam, 1986). Therefore, in this study, we applied subjective measures to assess the FP of SMEs. We have defined FP as the superiority that a firm gained compared with its competitors through implementing cloud computing in the operations and processes. Extant research suggests that perceived competitive performance, such as profitability, growth and market share, is a good indicator of business superiority (Lee *et al.*, 2016; Felipe *et al.*, 2019; Singh *et al.*, 2016).

3. Research framework and hypotheses development

We have developed our research model by integrating two prominent theoretical frameworks, namely, UTAUT and TOE. UTAUT is originated based on eight earlier theoretical models of IT adoption and is considered as the most predominant and comprehensive model for investigating the determinants of innovative technology adoption.

Scholars have applied UTAUT extensively in their studies to understand technology adoption process (Dwivedi *et al.*, 2017; Talukder *et al.*, 2018b; Ooi *et al.*, 2018). It consists of four fundamental constructs, namely, performance expectancy (PE), effort expectancy (EE), social influence and facilitating conditions to explain user acceptance and use behaviour of any new technology (Venkatesh *et al.*, 2003). This unified model explains up to 70 per cent variance of the behavioural intention to accept new technology (Teo *et al.*, 2015; Talukder *et al.*, 2018a). Also, UTAUT acknowledges the influence of external factors such as social factors (Zuiderwijk *et al.*, 2015) and can examine complicated and sophisticated organisational, technology and managerial issues (Venkatesh *et al.*, 2003). Further, Venkatesh *et al.* (2012) extended UTAUT by introducing some new constructs (hedonic motivation, price value and habit). They also incorporated moderating variables such as age, gender and experience to better explain the adoption of technology. However, in this study, we have adopted PE and EE as two key predictive constructs of cloud services adoption by SMEs because these two constructs fit in organisation context (Ooi *et al.*, 2018).

Similarly, Tornatzky *et al.* (1990) proposed the TOE framework based on Fred Fiedler's contingency theory that groups the common factors that affect technology adoption at organisation context. TOE has been considered as one of the popular frameworks that focus on people rather on roles and believes that people are not a replaceable component (Awa *et al.*, 2017). The key components of this framework are technological conditions, an organisation's internal conditions and external industry environment. Moreover, this framework is organisation size and industry restrictions free (Gangwar *et al.*, 2015). Previous scholars have recurrently worked on the factors within TOE's framework to reinforce its theoretical ground and have proved theoretical and empirical conformity (Awa *et al.*, 2015). Therefore, TOE offers a comprehensive view of the adoption of technology, its implementation and the post-adoption diffusion among organisations (Oliveira *et al.*, 2014; Gupta *et al.*, 2013; Makena, 2013).

In our study, PE and EE were considered as technological factors. Besides, this study adopted absorptive capacity (AC) and TR as organisational factors, and competitive pressure (CP) and regulatory support (RS) as environmental factors. Further, the inclusion of additional context-related constructs allows a more explicit examination of a user's acceptance of specific technology (Cimperman *et al.*, 2016). Therefore, this study added data security and privacy (DSP) and perceived trust (PTR) in the proposed model. The presence of DSP concerns towards using technology is significant for enterprises because security and privacy are considered as major challenges for adopting cloud computing (Oliveira *et al.*, 2014). Trust towards cloud providers and cloud technology itself plays an essential role in adopting cloud services by organisations (Alkhater *et al.*, 2018). Furthermore, this study attempts to assess the post-adoption impact of cloud services in SMEs. We assessed if the utilisation of cloud services improves the FP through enhancing the ORGA. To do this, we included ORGA as a mediator and FP as the second key dependent construct in the proposed model.

Finally, three control variables (e.g. firm size, firm age and industry) have been added to reduce the variance in the FP that might be affected by these variables. Higher financial and labour resources owned by large organisations may lead to variances in FP (Ilmudeen and Bao, 2018). Firm age would affect the performance of a firm because older firms may have sustainable inter-firm relationships, bargaining power in the current market and the popularity of internal routines (Ravichandran *et al.*, 2005). The industry is included as a control variable to address the performance variances that might be occurred due to industry-specific physiognomies such as industry environment and market competition intensity (Oliveira *et al.*, 2014).

Based on the above discussion, we developed the research framework of the study (see Figure 1). Research hypotheses are explained in the following.

3.1 Performance expectancy

PE refers to the degree to which the use of a system offers benefits to organisations in executing different activities more smoothly in day-to-day work (Venkatesh *et al.*, 2012). Prior literature suggests that this factor has a strong influence on behavioural intention of technology adoption (Oliveira *et al.*, 2016; Slade *et al.*, 2015). In the domain of cloud computing, organisations believe that using cloud technology will help them perform operational tasks more conveniently and efficiently. Cloud technology offers several benefits for SMEs such as cost reduction, business' time saving, scalability, flexibility, mobility and shared resources. Thus, it is meaningful to examine whether these advantages affect PE and successively the adoption of cloud computing by SMEs:

H1. Performance expectancy has a positive influence on cloud computing adoption.

3.2 Effort expectancy

EE refers to the degree of convenience involved with the use of an IS (Venkatesh *et al.*, 2012). The simplicity in using an innovative technology plays a weighty role in adopting that IS. In the SME context, EE indicates how much endeavour an enterprise needs to give to learning and using innovative technology. The possibility of adopting new technology will be higher if users feel ease to integrate the technology into organisational activities (Oliveira *et al.*, 2014). As SMEs have a lower technical workforce to use advanced and complicated technology, they perceive the use of cloud services to be free of effort. In Bangladeshi context, cloud computing is fairly a new concept to SMEs; therefore, it is imperative to discover whether Bangladeshi SMEs observe cloud services easy or tough and if this perception affects the adoption decision of cloud services:

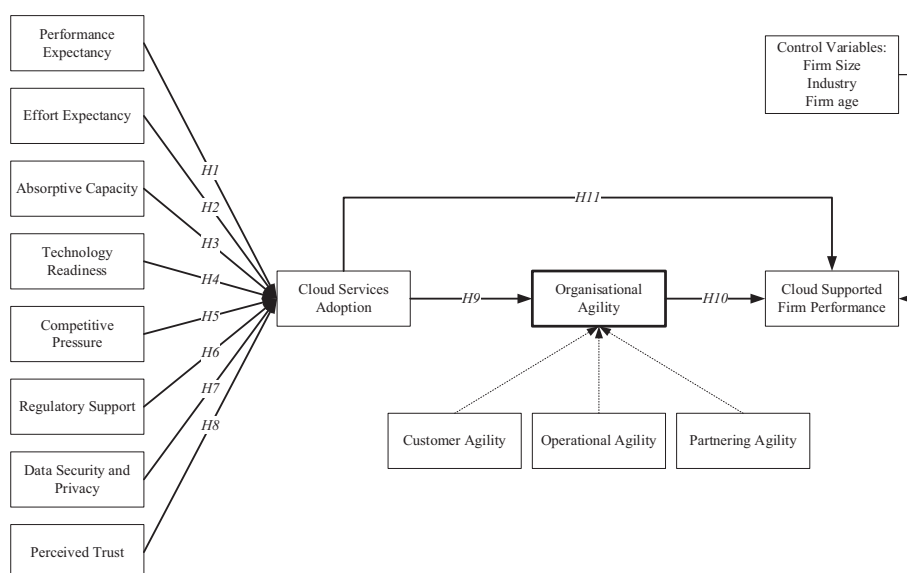


Figure 1.
Proposed research
model

H2. Effort expectancy has a positive effect on cloud computing adoption.

3.3 Absorptive capacity

Cohen and Levinthal (1990) defined AC as “the firm’s ability to recognise the value of new information, assimilate it, and apply it to commercial ends”. Zahra and George (2002) further defined AC as “a set of organisational routines and processes by which firms acquire, assimilate, transform and exploit knowledge to produce a dynamic organisational capability”. The level of AC of a firm is determined by the previous related knowledge and intensity of effort. Particularly, in a dynamic environment, AC can form the expectations and allows a firm to more accurately foresee the nature and potential of emerging technologies (Wei *et al.*, 2015). According to Todorova and Durisin (2007), the higher level of AC helps firms in recognising the value of innovative external knowledge, and acquire, assimilate, transform and exploit that knowledge. As cloud computing is treated as a disruptive technology, firms with a high level of AC can predict its value and assimilate it to firms’ processes and operations (Ooi *et al.*, 2018).

Previous literature suggests that a firm is likely to adopt new technology if it has relevant skills and knowledge regarding similar technologies (Lian *et al.*, 2014; Cohen and Levinthal, 1990). For example, Lian *et al.* (2014) state that if a firm has the knowledge and skills of implementing electronic data interchange (EDI) technology, it is likely to adopt radio frequency identification (RFID). Therefore, we anticipate that firms that accumulated rich technical and managerial experiences (AC) in using network-based collaborative technologies (e.g. e-SCM) are likely to adopt cloud computing services. Accordingly, we hypothesise the following:

H3. Absorptive capacity positively influences cloud computing adoption.

3.4 Technology readiness

The TR includes both the structural aspects and the technical human resources of an organisation. The structural aspects refer to the technical infrastructure such as installed network technologies and enterprise systems. On the other hand, the technical human resources refer to the workforce within the organisation who have IT knowledge and skill to implement the cloud-computing services. A number of empirical studies recommended that TR affect technology adoption significantly (Oliveira *et al.*, 2014; Makena, 2013). In SMEs, TR such as the existence of the IT infrastructure and IT skills plays a significant role in an organisation’s decision to use cloud technology. A strong internet connection, high bandwidth and high internet speed, along with skilled human resources, make it convenient to implement cloud computing in SMEs. Hence, the following hypothesis is postulated:

H4. Technology readiness has a positive effect on the adoption of cloud computing.

3.5 Competitive pressures

As the business environment is dynamic, every enterprise rigorously screens other player’s actions to achieve competitive advantages to sustain in the market. New technology adoption is usually recognised as a strategic stipulation for firms to survive in today’s tremendous competitive and continuously changing business environment. In the innovation diffusion literature, scholars have found CPs as an essential factor that affects

technology adoption process. Oliveira and Martins (2011) stated that enterprises mimic different personas of successful competitors, particularly those that are related to new technology adoption. However, with the adoption of cloud computing, SMEs might improve their business operations, enhance market efficiencies, protect data and, thus, can develop their competitiveness. Therefore, we postulate the following hypothesis:

- H5. The existence of competitive pressures positively influences cloud computing adoption.

3.6 Regulatory support

Support from regulatory bodies is an essential factor that affects the diffusion of innovation (Zhu and Kraemer, 2005; Zhu *et al.*, 2006). RS is defined as the necessary support provided by a government agency to facilitate the diffusion of IT innovation among firms (Zhu *et al.*, 2006). A government can affect the diffusion of innovation either by regulating tax and other measures to increase or decrease payoff or by altering the climate in which innovation is received (Williamson, 1980). According to Zhu *et al.* (2006), different supportive regulations and policies of government motivate e-business assimilation. Oliveira *et al.* (2014) state that governmental RSs can encourage or discourage firms from adopting cloud services. For instance, in the USA and the European Union, members have explicit directives to protect organisational data (Oliveira *et al.*, 2014). If government of a country has specific standards and protocols for cloud computing, firms will be more enthusiastic to accept cloud computing. Hence, the following hypothesis is posited:

- H6. Regulatory support positively affects cloud computing adoption.

3.7 Data security and privacy

The degree of measures that a cloud provider undertakes to secure its cloud system or protect customer firm's data resources from any unwanted attacks or threats is called security (Lian *et al.*, 2014). A security breach occurs when a firm or a government agency loses information, personal records or other sensitive data (Oliveira *et al.*, 2014). Cloud computing converges storage and computing to offer a shared environment which raises the security issues. Another severe challenge of cloud computing is to manage the identity in the cloud environment. If the cloud provider fails to develop mature security protocols and identity management, the customer firms will be reluctant to accept cloud services (Oliveira *et al.*, 2014). In terms of privacy, firms that use cloud services lose their full control over their stored information in the server owned and run by a third party (Alkhater *et al.*, 2018). This issue raises the degree of risk regarding data confidentiality and might direct to a privacy breach (Takabi *et al.*, 2010). Security and privacy concerns significantly discourage firms towards the adoption of cloud services. Therefore, we develop the following hypothesis:

- H7. Data security and privacy concerns will have a negative effect on cloud computing adoption.

3.8 Perceived trust

Singh and Sirdeshmukh (2000) defined trust as "the psychological state leading to accepting the vulnerability of a trustor, based on positive expectations of the trustee's actions". The implication of trust is emphasised in the cloud computing context because of the

longitudinal and chronological separation between cloud providers and users. The lack of transparency and control over data which are stored in the cloud could cause trust problems (Alkhater *et al.*, 2018). Trust includes not only trusting cloud providers but also trusting the technology itself. This research assumes that the more the organisation has the willingness and confidence in storing its resources in a third party's cloud, the more they will adopt cloud computing:

H8. Perceived trust has a positive impact on cloud computing adoption.

3.9 Organisational agility

IT capabilities play a vital role in developing ORGA by enhancing a firm's capacity to sense and respond to a dynamic environment (Mikalef and Pateli, 2017). Firms develop a set of IT capabilities consisting of digitised work processes and knowledge management systems with the integration of IT with the key business process to strengthen innovation and build digital options (Felipe *et al.*, 2019), which enable a firm to improve ORGA (Sambamurthy *et al.*, 2003). Nowadays, organisations need to handle a huge volume of information collected from the environment, as well as from its own internal processes. Also, they must analyse and leverage this information to provide a quick response to face their challenges. Cloud computing, as prevalent information technology, serves as enablers for a firm to develop its IT capabilities, which subsequently help to increase a firm's agility. According to Oliveira *et al.* (2014), the proper implementation of cloud computing can help converging IT efficiency and business agility of the firms (Oliveira *et al.*, 2014). The ways of achieving IT efficiency include using of scalable IT resources (Marston *et al.*, 2011), improving efficiency in work (Low *et al.*, 2011) and ensuring the high availability of services (Armbrust *et al.*, 2010). On the other hand, the business agility of cloud computing includes the capacity of deploying mass computing technology quickly, minimising capital cost directly and responding rapidly to the market change (Oliveira *et al.*, 2014). In the age of the Industry 4.0, which has brought digitisation and automation in production and operation, the adoption and use of cloud computing offer more agility for the firms (Ooi *et al.*, 2018). In sum, the proper implementation and deployment of cloud technology can attain a higher level of ORGA by sensing and responding to environmental changes, scanning internal information, increasing innovativeness and making fast amendment in internal processes. Therefore, we hypothesise the following:

H9. Cloud computing implementation has a positive impact on organisational agility.

In the dynamic environment, ORGA contributes significantly to the growth and performance of the firm by improving value creation and value capture. ORGA enables a firm to sense, seize and transform capabilities, which plays an essential role in facing challenges arisen from the customers, suppliers, competitors and technological change (Teece *et al.*, 2016). Sambamurthy *et al.* (2003) stated that FP could be improved continuously by the extent of ORGA which brings innovations in product, process, relation, channels and market segmentation. Robust customer agility helps firms to respond proactively to customer demands by offering innovative products, promotions and services, which increase firms' revenue, competitive position and competitive advantage (Roberts and Grover, 2012). Similarly, operational agility plays a critical role in firm's competitive performance through enhancing flexibility in operations, reducing costs and amplifying customer retention (Mikalef and Pateli, 2017). Finally, FP is impacted by partnering agility through developing a strategic network and building trusted virtual

relationship with partners ([Sambamurthy et al., 2003](#)). Based on the above discussion, the following hypothesis is posited:

H10. Organisational agility will have a positive impact on firm performance.

3.10 Firm performance

In the domain of cloud technology, performance indicates the potentials of cloud services that can positively affect the cloud-supported operations and processes of an organisation. Cloud computing enables SMEs to concentrate more on core business capabilities, enhanced operational performance, productivity and financial performance. Earlier researches on cloud computing confirmed the positive effect of the implementation of cloud services on FP. For instance, [Ooi et al. \(2018\)](#) conducted a study in the context of Malaysian manufacturing companies and found a positive impact of cloud computing on non-financial performance (e. g. flexibility and quality). Similarly, [Garrison et al. \(2015\)](#) stated that the successful implementation of cloud computing positively influences the FP for cloud-supported operations. Therefore, we formulate the following hypothesis:

H11. The adoption of cloud computing has a positive impact on firm performance.

4. Methodology

4.1 Variables and measurements

A survey instrument was designed based on the consultation with an expert group of five professionals. The first segment of the instrument contained 46 close-ended questions, which denoted the measurement items of the constructs used in the research model. All the items were taken from the literature, and some of them were modified according to the research context (see [Appendix 1](#)). We used five-point Likert scale ranging from 1 to 5, where 1 = strongly disagree and 5 = strongly agree, to design the closed-ended questions. The second segment concerned about demographic information. We conducted a pilot test using 30 respondents, who were asked to participate voluntarily. We made some minor revisions of the questionnaire based on the results of pilot test. In response to some respondents' claims, we changed relatively infrequent words with the most common words on the condition that such change would not influence construct's content validity.

4.2 Sampling and data collection

The research model was tested with the data collected from SMEs in Bangladesh. The survey was administered among SMEs located in different market places, shopping malls, industrial area and export processing zones. While distributing questionnaires, we requested that the questionnaire be filled up by the owners or managers, who play key roles in decision-making process. This study selected respondents having relevant knowledge and willingness to respond. We applied a non-probabilistic convenience sampling technique as it was quicker and easier to manage compared to the other techniques that needed more resources (e.g. technical, financial and time). We distributed a total of 500 questionnaires to the respondents; among them, 411 questionnaires were returned resulting in 82.2 per cent response rate. We gathered data in the months from July to December 2018. After conducting data-cleansing process, a total of 372 questionnaires were assembled. The demographics of the participants and firms are presented in [Appendix 2](#).

Table I.
One-sample
Kolmogorov–
Smirnov test

4.3 Data analysis

We performed some statistical tests to assess multivariate assumptions. First, Kolmogorov–Smirnov’s test was conducted to check the normality of the dataset. The results of this test, as presented in [Table I](#), revealed that all *p*-values are lower than 0.05, which concluded that the dataset is not normally distributed. We also conducted Pearson’s skewness and Kurtosis. The results of this test indicated that all the values were in the acceptable range (in between -2.58 and $+2.58$).

Next, we checked the multicollinearity problem by using correlation coefficients and variance inflation factor (VIF) values. The results, as shown in [Table II](#), indicated that all the values for correlation coefficients are lower than the threshold of 0.90 ([Hair et al., 2006](#)). Besides, the VIF values ([Table II](#)) for the constructs were lower than the threshold of 10 ([O’Brien, 2007](#)). Thus, we can rule out the multicollinearity problem.

Finally, scatter plot was used to examine the homoscedasticity of distribution of the data ([Teo et al., 2015](#)). Based on the distribution of the regression standardised residuals, this study observed that all residuals are evenly scattered along a straight line, which indicates that the homoscedasticity of the distribution is supported.

When the multivariate assumptions were satisfied, we applied partial least square structural equation modelling (PLS-SEM) to confirm the reliability and validity of the constructs and to evaluate the predictive relevance of the model and the total variance explained ([Chin, 1998](#)). Next, we tested the relationships which are developed from the

Indicators	<i>N</i>	Normal parameters ^{a,b}		Most extreme differences				Test statistic	Asymp. sig. (two-tailed)
		Mean	SD	Absolute	Positive	Negative			
PAT1	372	3.45	0.85	0.227	0.216	−0.227	0.227		0.000
PAT2	372	3.44	0.862	0.272	0.187	−0.272	0.272		0.000
PAT3	372	3.20	0.746	0.258	0.258	−0.25	0.258		0.000
PAT4	372	3.48	0.897	0.241	0.182	−0.241	0.241		0.000
OP1	372	3.25	0.701	0.27	0.27	−0.257	0.27		0.000
OP2	372	3.17	0.696	0.287	0.283	−0.287	0.287		0.000
OP3	372	3.21	0.769	0.269	0.25	−0.269	0.269		0.000
CUS1	372	3.10	0.798	0.242	0.226	−0.242	0.242		0.000
CUS2	372	3.05	0.867	0.244	0.197	−0.244	0.244		0.000
CUS3	372	3.06	0.825	0.262	0.249	−0.262	0.262		0.000

Notes: ^aTest distribution is normal; ^bCalculated from data

Table II.
Correlation
coefficients, VIF and
discriminant validity

Constructs	AC	CP	DSP	EE	RS	PE	PTR	TR	VIF
AC	0.830								2.279
CP	0.589	0.857							2.824
DSP	0.550	0.660	0.814						2.621
EE	0.616	0.650	0.673	0.828					2.580
RS	0.022	0.026	0.037	0.003	0.916				1.006
PE	0.655	0.735	0.734	0.707	0.027	0.840			3.097
PTR	0.712	0.711	0.645	0.701	0.016	0.746	0.837		2.346
TR	0.562	0.654	0.623	0.575	0.063	0.628	0.603	0.828	2.105

research model. Also, we have conducted IPMA in PLS-SEM to recognise the critical factors of cloud computing adoption and competitive performance. The PLS-SEM approach was considered as the most appropriate approach in this study as this method can examine a series of relationships simultaneously, whereas other statistical methods (e.g. multiple regression analysis) are confined to analysing the relationships between each construct singly (Hair *et al.*, 2016). The confirmatory approach of SEM rather than descriptive supersedes other multivariate approaches in testing a theoretical model. Besides, the nature of this study better suits PLS-SEM as this research focuses on theory development or theory exploration rather than theory confirmation (Tan and Ooi, 2018). Also, the complexity of the model and a large number of indicators lead to the use of variance-based PLS-SEM (Ooi *et al.*, 2018). Finally, as this study aims to enhance the variance explained, which is data-driven rather than estimation of model fit, we used variance-based PLS-SEM in place of covariance-based SEM (Tan *et al.*, 2018).

4.4 Non-response bias

Similar to Ooi *et al.* (2018), we assessed the differences between all key variables using the independent *t*-test and the result indicated that there are no significant differences identified. For further verification, we conducted Chi-squared test for independence, and the result showed that there are no significant differences. Thus, non-response bias was not found as a potential problem in this study.

4.5 Common method bias

Following the recommendations given by Schwarz *et al.* (2017), we endeavoured to control common method bias (CMB) problem during the research design phase. To do this, we applied both procedural remedies and statistical tests. Procedurally, similar to Ooi *et al.* (2018), we confirmed respondents that their identities and responses would be anonymous. Further, we motivated them so that they respond unequivocally. Also, we focused more on the wording of the items to reduce ambiguity. Besides, we performed some statistical tests to detect CMB. First, we applied Harman's single-factor approach. To identify the number of factors critical for the description of variance, we applied principal axis factor analysis (PAF) (Harman, 1976). The results of the test suggested that a single construct was responsible for 39.08 per cent of the total variance, which is far below than the recommended 50 per cent (Podsakoff *et al.*, 2003). Also, we examined the correlation matrix of the constructs (Table II) and found that there was no correlation over 0.90 (Bagozzi *et al.*, 1991). Finally, we tested CMB based on VIF values (Table II) of the constructs, which were less than the suggested value of 3.3 (Kock, 2015). Thus, CMB problem was not a serious problem in this study.

5. Results

5.1 Measurement model

To examine the measurement model, we applied internal reliability, convergent validity and discriminant validity criteria (Hair *et al.*, 2013).

5.1.1 Assessment of construct reliability. We evaluated the reliability of the constructs by using Cronbach's alpha, Dijkstra-Henseler's rho (ρ_A) and composite reliability. The value of reliability criteria (as shown in Table III) was greater than the suggested value of 0.70 (Henseler *et al.*, 2009). The reliability of items was measured based on item loadings, which were greater than recommended 0.70.

5.1.2 Convergent validity and discriminant validity. We evaluated convergent validity by testing the values of average variance extracted (AVE). It is evident from the results, as

Constructs	Items	Scale type	Loadings/ weights ^a	<i>t</i> -value	Cronbach's alpha	rho_A	CR ^b	AVE ^c
<i>First-order constructs</i>								
Performance expectancy (PE)	PE1	Reflective	0.823	38.171**	0.861	0.868	0.906	0.706
	PE2		0.863	55.338**				
	PE3		0.866	65.135**				
	PE4		0.807	36.470**				
Effort expectancy (EE)	EE1	Reflective	0.818	42.822**	0.847	0.850	0.897	0.686
	EE2		0.820	42.612**				
	EE3		0.831	47.419**				
	EE4		0.843	64.942**				
Absorptive capacity (AC)	AC1	Reflective	0.877	61.581**	0.860	0.854	0.901	0.720
	AC2		0.902	76.089**				
	AC3		0.873	40.427**				
	AC4		0.829	36.284**				
Technology readiness (TR)	TR1	Reflective	0.769	24.498**	0.771	0.787	0.867	0.686
	TR2		0.870	67.289**				
	TR3		0.842	43.088**				
Competitive pressure (CP)	CP1	Reflective	0.829	41.997**	0.819	0.821	0.892	0.734
	CP2		0.865	59.519**				
	CP3		0.876	77.220**				
Regulatory support (RS)	RS1	Reflective	0.926	53.643**	0.909	0.932	0.940	0.840
	RS2		0.939	57.569**				
	RS3		0.884	51.209**				
Data security and privacy (DSP)	DSP1	Reflective	0.805	40.449**	0.821	0.846	0.877	0.662
	DSP2		0.814	43.605**				
	DSP3		0.798	29.914**				
	DSP4		0.840	54.313**				
Perceived trust (PTR)	PTR1	Reflective	0.887	86.579**	0.857	0.862	0.903	0.701
	PTR2		0.853	57.116**				
	PTR3		0.803	42.681**				
	PTR4		0.803	43.012**				
Cloud adoption (CA)	CA1	Reflective	0.893	71.354**	0.866	0.866	0.918	0.789
	CA2		0.906	95.472**				
	CA3		0.864	57.781**				
Firm performance (FP)	FP1	Reflective	0.725	21.167**	0.791	0.806	0.865	0.616
	FP2		0.849	58.197**				
	FP3		0.817	40.010**				
	FP4		0.741	26.386**				
<i>Second-order constructs</i>								
Operational agility (OPA)	OPA1	Formative	0.482	27.091**	N/A	N/A	N/A	N/A
	OPA2		0.465	25.821**				
	OPA3		0.256	13.437**				
Customer agility (CUSA)	CUS1	Formative	0.240	22.718**	N/A	N/A	N/A	N/A
	CUS2		0.410	44.169**				
	CUS3		0.503	56.152**				
Partnering agility (PARTA)	PART1	Formative	0.181	12.448**	N/A	N/A	N/A	N/A
	PART2		0.396	25.721**				
	PART3		0.233	13.512**				
	PART4		0.375	23.901**				

Table III.
Measurement model

Notes: **Significant at $p < 0.001$; ^aLoadings are for reflective items and weights are for formative items; ^bCR = Composite reliability, ^cAVE = Average variance extracted

depicted in Table III, that the values of AVE were above the recommended values of 0.50 (Fornell and Larcker, 1981). Also, all factor loadings (>0.50) (see Table III) and composite reliability (>0.70) (see Table III) confirmed the convergent validity of the constructs (Henseler *et al.*, 2009). Likewise, we examined discriminant validity by checking the correlations among the measures of possibly overlying variables. In this paper, the value of square root of AVE for each variable was above the correlation of that variable with other variables (see Table II). Additionally, all values of heterotrait–monotrait ratio ratio, as shown in Table IV, were <0.90 . These measures confirmed the discriminant validity of the studied constructs (Henseler *et al.*, 2014).

5.1.3 Quality assessment of second-order formative construct. We applied the approach offered by Petter *et al.* (2007) for the assessment of the quality of the second-order formative constructs. First, we examined multicollinearity considering VIF of the formative items of ORGA. All VIF values, as depicted in Table V, were lower than 10, which confirmed that there is no multicollinearity issue. Further, we assessed the weights of indicators. Table III demonstrates that all indicators' weights for ORGA were strongly significant. Finally, the significance of the first-order constructs (customer agility, operational agility and partner agility) which are considered second-order construct's (ORGA) indicators was tested and found that all criteria are satisfied (see Table VI).

5.2 Structural model

We performed bootstrap (5,000 re-samples) using SmartPLS 3.2.8 version software. The results indicated that the model has the ability to explain 76.0 per cent variance in cloud adoption, 48.6 per cent variance in ORGA and 50.9 per cent variance in FP. This study examined the hypotheses using path coefficient (β) and *t*-statistics. The results of all hypotheses testing are depicted in Figure 2. Appendix 3 contains the detailed results of hypotheses testing.

Constructs	AC	CP	DSP	EE	RS	PE	PTR	TR
AC								
CP	0.705							
DSP	0.651	0.798						
EE	0.723	0.776	0.797					
RS	0.034	0.040	0.060	0.023				
PE	0.760	0.841	0.834	0.827	0.034			
PTR	0.833	0.848	0.760	0.822	0.034	0.828		
TR	0.685	0.820	0.777	0.701	0.073	0.765	0.732	

Table IV.
Heterotrait–
monotrait ratio
(HTMT)

Coefficients ^a			
Model	Collinearity statistics		VIF
	Tolerance		
<i>I</i>			
CUSA	0.643		1.556
OPA	0.624		1.603
PARTA	0.564		1.772

Note: ^aDependent variable: ORGA

Table V.
Multicollinearity test

The results indicated that hypotheses *H1*, *H2*, *H3*, *H7*, *H8*, *H9*, *H10* and *H11* were statistically significant. This means that the hypothesised relationships of PE; EE; AC; DSP; and PTR with cloud computing adoption are confirmed. The adoption of cloud computing positively influences ORGA; subsequently, ORGA strongly affects FP. The direct relationship of cloud computing adoption with FP is also supported. In contrast, the *p*-values (>0.05) for the paths for hypotheses *H4*, *H5* and *H6* indicated that TR, CP and RS did not affect the adoption of cloud computing. The influence of three control variables (e.g. firm size, firm age and industry) was found statistically insignificant, which indicated that these variables did not cause any significant variance in FP.

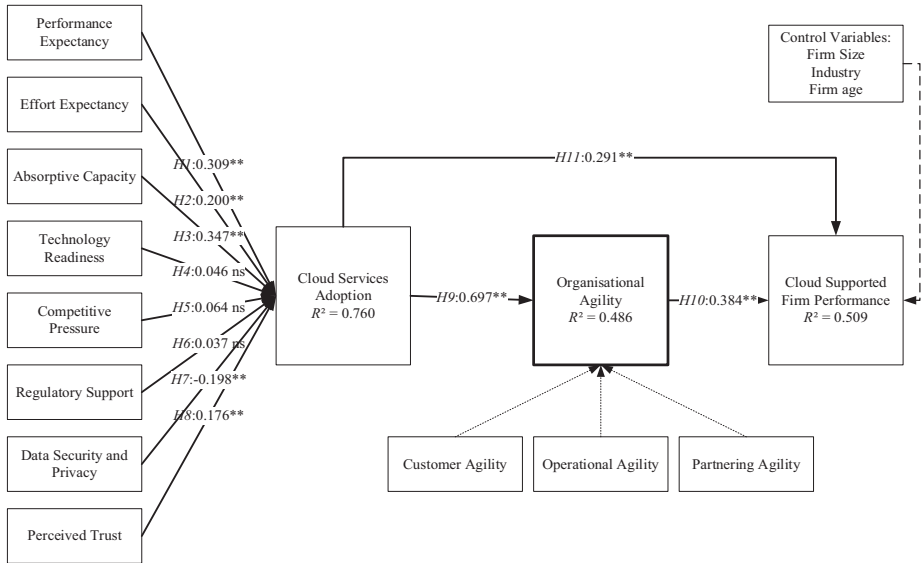
5.3 Predictive relevance and the effect size

We applied Stone–Geisser’s Q^2 to examine predictive relevance of the endogenous constructs. The Q^2 values of 0.02, 0.15 and 0.35 indicate a low, moderate and large amount of predictive relevance, respectively (Cohen, 2013). The Q^2 values for endogenous constructs (as shown in Table VII) are all positive and greater than 0.35. The results imply that all endogenous constructs have significant predictive relevance. Also, the R^2 values determine the predictive capability of these endogenous constructs. The R^2 values for cloud computing

Table VI.
Weight of first-order
constructs

Second-order construct/path	Weight	<i>t</i> -value
Organisational agility (ORGA)		
CUSA → ORGA	0.343	12.075
OPA → ORGA	0.333	11.466
PARTA → ORGA	0.502	18.229

Figure 2.
Structural model with
standardised beta
values and explained
variance



Notes: $*p < 0.05$; $**p < 0.01$; ns = Not supported

adoption, ORGA and FP are 0.760, 0.486 and 0.509, respectively. These indicated that these constructs possess acceptable predictive capacity. In addition, we assessed the effect size by using the f^2 values for cloud adoption and ORGA on FP. The f^2 value (Table VII) indicated that the effect size of cloud adoption and ORGA on FP is significant (>0.15).

5.4 Importance–performance map analysis

According to Ringle and Sarstedt (2016):

The IPMA gives researchers the opportunity to enrich their PLS-SEM analysis and, thereby, gain additional results and findings. The goal of running IPMA is to distinguish factors that have a relatively high total effect (importance) but relatively low performance in shaping endogenous variables (Hair *et al.*, 2016).

Thus, the IPMA allows observing those areas that are of superior importance to a target construct even though the construct is performing poorly (Nitzl and Chin, 2017), thereby signifying major areas to emphasise or improve.

In this study, we ran IPMA in PLS using SmartPLS 3.2.8 version software for several times. First, we performed IPMA taking cloud adoption as the target construct.

From Figure 3, we can observe that PE and AC have both higher importance score and performance score. This result indicates that firms are pretty well in managing these two factors. Also, PTR and DSP are effectively controlled by the firms as they have moderate importance score and performance score. In contrast, firms have overemphasised TR, CP and RS as they have very low importance score in shaping cloud adoption but higher performance score. Essentially, EE is the area that should be well monitored by the firms as it has lower performance score compared to other constructs in spite of its high importance in explaining cloud adoption.

Endogenous constructs	R^2	Q^2	f^2
CA	0.760	0.729	0.160
ORGA	0.486	0.475	0.154
FP	0.509	0.497	

Table VII.
Predictive relevance
and the effect size

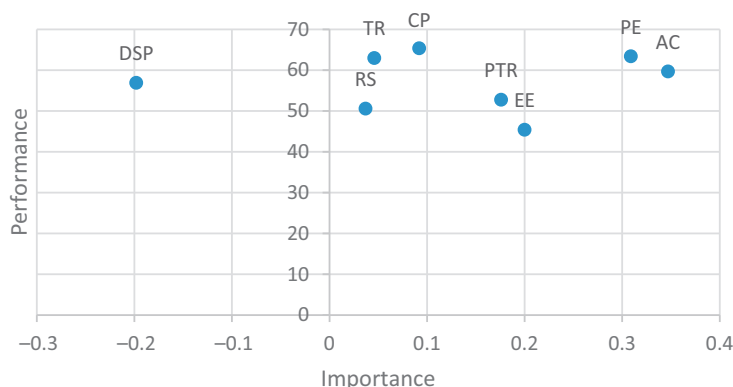


Figure 3.
Importance–
performance map
(using cloud adoption
as target construct)

Second, we run IPMA for assessing two direct predecessors (e.g. ORGA and cloud adoption) of FP. We can see from Figure 4 that managers are quite efficient in managing cloud adoption and ORGA areas. Among these two, cloud adoption is more important for predicting FP. Therefore, managers should emphasise more on this issue.

Finally, we ran IPMA to measure the three dimensions (e.g. customer agility, operational agility and partnering agility) that form the independent second-order construct (e.g. ORGA). From Figure 5, it can be observed that all three dimensions are placed within the upper right quadrant. This outcome indicates that all these dimensions' score over the average both in terms of importance and performance, and thus, they must be carefully controlled or monitored. Among them, the most important dimension seems to be partnering agility and operational agility; both of them score well above the average in terms of importance. In addition, the customer agility dimension scores over the average in terms of importance, yet its distance from the average is more reduced.

6. Discussion

This study found that PE and EE, two UTAUT constructs, significantly influenced cloud computing adoption in SMEs. The findings are supported by several former studies (Alalwan *et al.*, 2017; Martins *et al.*, 2014). This study confirms that users' (e.g. managers,

Figure 4.
Importance–
performance map
(using firm
performance as target
construct)

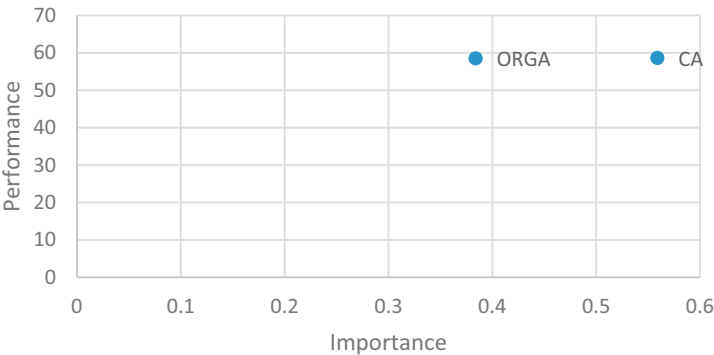
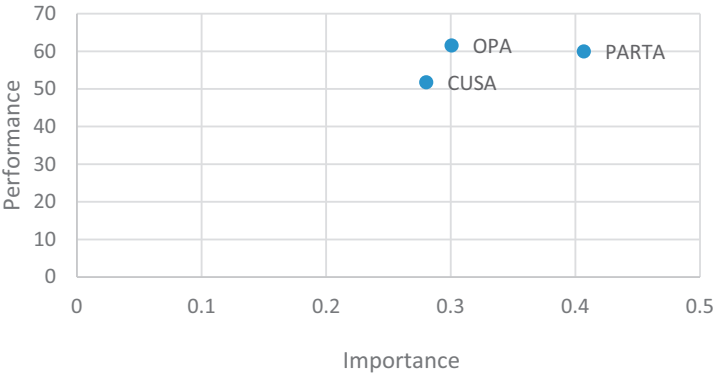


Figure 5.
Importance–
performance map
(using organisational
agility as target
construct)



owners, employees) characteristics are important for cloud computing assimilation in SMEs. Users in SMEs always expect that the utilisation of cloud will improve their performance. Also, they expect a simple technology that requires fewer efforts, minimum technical knowledge and less time.

This study identified that the level of AC of a firm significantly motivated to adopt an emerging technology such as cloud computing. The finding is consistent with [Wei et al. \(2015\)](#), who claim that AC positively influences RFID assimilation in firms in the context of China. This finding implies that a firm with the capacity of sensing, acquiring, assimilating and exploiting external knowledge has the high possibility of adopting cloud computing in its operations and processes. In contrast, this study revealed that TR had an indirect effect on the adoption of cloud services. This result contradicts some prior studies on technology adoption ([Senyo et al., 2016](#); [Gutierrez et al., 2015](#); [Oliveira et al., 2014](#)). This result advises that lack of TR impedes enterprises from using cloud technologies. One reason is the unsatisfactory technical infrastructure in Bangladesh. The unstable and slow internet along with the high cost of using internet services hinders the penetration of network-based services in Bangladesh ([Hoque et al., 2016](#)). Moreover, SMEs suffer from the lack of technological infrastructure and knowledge required for implementing cloud computing.

Consistent with [Oliveira et al. \(2014\)](#), this article identified that CP had an indirect influence on cloud adoption in SMEs in Bangladesh. This finding implies that the extent of pressure that SMEs are facing is still insignificant. Majority of Bangladeshi firms are still unable to convert the benefits of using cloud technologies into their competitive advantages. Similarly, RS was also identified as an insignificant predictor of cloud computing adoption, particularly in Bangladesh. This finding does not indicate that SMEs disrespect existing laws and regulations, rather existing standards and policies do not encourage SMEs to accept cloud technology. RSs are crucial to inspire the sense of trust needed for SMEs to transform innovation into business opportunities ([Oliveira et al., 2014](#)). The government should enforce cloud-supported standards and regulations, and, at the same time, offer financial incentives to remove the impediments of cloud computing adoption.

Concern towards DSP was found as a significant inhibitor of cloud adoption in this research context. This result is consistent with some former researches conducted on the acceptance of innovative technology ([Johnson et al., 2018](#); [Yang et al., 2013](#)). SMEs are very anxious about the safety of their resources that are stored in the third party's server. Also, this study found a significant effect of trust on the adoption of cloud services in SMEs. This finding is supported by a number of previous studies which claimed trust as an essential prerequisite for user IT adoption ([Shareef et al., 2011](#); [Hampshire, 2017](#); [Sharma et al., 2016](#); [Alkhater et al., 2018](#)).

Further, this study identified that cloud computing implementation and ORGA is positively related, and further, ORGA strongly influenced FP. This outcome implies that cloud computing adoption enhances SMEs' competitive performance through improving ORGA in the form of customer agility, operational agility and partnering agility. Finally, this study established a positive relationship between cloud implementation and firm's comparative performance. This finding is not astonishing because organisations migrate to the cloud to save cost and achieve performance ([Garrison et al., 2015](#)). With the successful implementation of cloud computing, SMEs can improve their non-financial performance by enhancing market share, the quality of product or service and corporate growth, and financial performance by increasing revenue, profit margin, sales, ROA, ROE and ROI. This finding is supported by several previous studies on cloud computing ([Ooi et al., 2018](#); [Garrison et al., 2015](#)).

7. Implications and future research direction

7.1 Theoretical implications

This study has added some noteworthy contributions to cloud computing diffusion literature. In response to scholars' recent calls, we have combined multiple theoretical models (e.g. UTAUT and TOE framework) to provide deeper understandings about the underlying impact of cloud computing adoption, ORGA and FP. Further, this study has examined several important relevant contextual constructs (e.g. DSP, PTR), which are not well explored in the cloud computing literature. Furthermore, the model has been extended with ORGA and FP to observe the effect of cloud computing adoption. Few studies in the literature perform such kind of studies.

This study has claimed that the blended model has a higher explanatory capacity compared to the predictability of any single model. The results of this study recommend that the proposed model has significant predictive capacity as the model has explained 76 per cent of the variance of cloud computing adoption. This study offers not only an integrated model in the context of cloud computing but also the generalisability; thus, this model can be used to assess other technology. We have tested the validity and reliability of the scales used in the survey instrument.

Finally, this study has performed IPMA in PLS to assess the factors based on their importance and performance scores. The outcomes of IPMA offer an empirical basis for improving the performance of a target construct (Ringle and Sarstedt, 2016). In this study, the outcomes of IPMA indicate that the PE and AC are efficiently managed and controlled. In contrast, managers need to focus more on EE, as it has a higher importance score but lower performance scores, for improving its performance towards shaping cloud adoption. In terms of FP, managers are efficient in monitoring cloud adoption and ORGA. Moreover, the IPMA outcomes denote that partnering agility and operational agility are the most important among the three dimensions that form ORGA, indicating that these dimensions should be carefully controlled by managers.

7.2 Practical implications

This study provides noteworthy insights for cloud providers and practitioners about the influential factors of cloud adoption. The outcomes of this research signify that PE and EE play crucial roles in adopting cloud computing by SMEs. Hence, cloud providers must consider users' requirements while developing cloud services. Also, they must offer user-friendly interfaces so that users with limited technical knowledge can easily interact with the cloud. Adequate security and privacy measures must be taken to save firms from any undesirable cases of cyber-fraud and security breaches. Cloud providers must establish a trustworthy environment with the customer firm by offering various benefits and services related to the utilisation of cloud computing. Overall, the outcomes of this research can improve the cloud providers' existing knowledge about why cloud adoption is still low in Bangladesh, and thus, they can make necessary efforts to foster the adoption of cloud services in SMEs.

SMEs should focus extensively on enhancing the AC, which will help them to assess the potentials of cloud computing very quickly. Also, the higher level of AC assists in bringing implementation success of cloud computing in SMEs. SMEs should focus on developing various organisational resources (e.g. technology infrastructure, IT knowledge, etc.) to gain the maximum benefit of cloud computing. Moreover, SMEs need various RS from the government agency to adopt cloud computing. The government of Bangladesh must develop cloud-friendly regulations to facilitate cloud adoption. For example, the government can introduce suitable cyber laws and standards to protect SMEs' information resource

stored in the cloud. Also, the government agency can undertake various incentive packages to motivate cloud adoption by SMEs. Finally, this study suggests that the implantation of cloud computing influences comparative FP directly and through improving ORGA. Therefore, the outcome of this research will motivate those managers who are still thinking about the adoption of cloud computing towards the implementation of cloud in their firms.

8. Future research direction

This study has some limitations that may lead to future lines of research. Although this study is conducted meticulously to confirm the validity of our research model, this study must acknowledge the likelihood that the findings may be subject to a specific research context. First, the sample was confined to a particular country such as Bangladesh, which indicates that this research reveals only the state of that particular country. Therefore, this research suggests conducting a further study using other similar country data. Second, data were collected from urban areas; firms of rural areas may not depend on cutting-edge technology. Thus, future research can be performed based on a larger sample collected from all parts of the country. Third, the proposed research model could be validated using longitudinal data. Finally, this study used linear statistical methods for analysing data; further research can be conducted using nonlinear and non-compensatory tools such as neural network, support vector machine, random forest and fuzzy set, which have the robust predictive capability.

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Constructs	Items	Adapted sources
Performance expectancy	Four items	Venkatesh <i>et al.</i> (2012), Ooi <i>et al.</i> (2018)
Effort expectancy	Four items	Venkatesh <i>et al.</i> (2012), Ooi <i>et al.</i> (2018)
Absorptive capacity	Four items	Ooi <i>et al.</i> (2018), Wei <i>et al.</i> (2015)
Technology readiness	Three items	Oliveira <i>et al.</i> (2014), Senyo <i>et al.</i> (2016)
Competitive pressure	Three items	Kumar <i>et al.</i> (2017), Low <i>et al.</i> (2011)
Regulatory support	Three items	Oliveira <i>et al.</i> (2014), Wei <i>et al.</i> (2015)
Perceived trust	Four items	Alkhater <i>et al.</i> (2018), Sharma <i>et al.</i> (2016)
Data security and privacy	Four items	Oliveira <i>et al.</i> (2014), Alkhater <i>et al.</i> (2018)
Cloud adoption	Three items	Gupta <i>et al.</i> (2013), Alkhater <i>et al.</i> (2018)
Operational agility	Three items	Lu and Ramamurthy (2011), Felipe <i>et al.</i> (2019)
Customer agility	Three items	Lu and Ramamurthy (2011), Felipe <i>et al.</i> (2019)
Partnering agility	Four items	Liu <i>et al.</i> (2016), Felipe <i>et al.</i> (2019)
Firm performance	Four items	Felipe <i>et al.</i> (2019), Lee and Choi (2003)

Table AI.
Measurement items

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Table AII.
The demographics of
the participants and
firms

Descriptions	Frequency	Percentage
<i>Respondents</i>		
Owner/CEO	211	56.72
IT manager	161	43.28
<i>Work experience (years)</i>		
0-2	42	11.29
3-5	65	17.47
6-10	120	32.26
More than 10	145	38.98
<i>Firm age</i>		
Less than 1	19	5.11
2-5	56	15.05
6-10	87	23.39
11-20	112	30.11
More than 20	98	26.34
<i>Firm size (Employees)</i>		
16-50	68	18.28
50-100	94	25.27
100-200	142	38.17
More than 200	68	18.28
<i>Industry type</i>		
Service	202	54.30
Manufacturing	170	45.70
<i>Market scope</i>		
Local	86	23.12
National	132	35.48
International	154	41.40

Hypotheses	Paths	Coefficient (β)	<i>t</i> -statistics	<i>p</i> Values	Decision
<i>H1</i>	PE → CA	0.309	5.431	0.000	Supported
<i>H2</i>	EE → CA	0.200	4.236	0.000	Supported
<i>H3</i>	AC → CA	0.347	6.452	0.000	Supported
<i>H4</i>	TR → CA	0.046	1.174	0.241	Not supported
<i>H5</i>	CP → CA	0.064	1.751	0.073	Not supported
<i>H6</i>	RS → CA	0.037	1.698	0.090	Not supported
<i>H7</i>	DSP → CA	-0.198	4.578	0.000	Supported
<i>H8</i>	PTR → CA	0.176	3.740	0.000	Supported
<i>H9</i>	CA → ORGA	0.697	22.470	0.000	Supported
<i>H10</i>	ORGA → FP	0.384	7.166	0.000	Supported
<i>H11</i>	CA → FP	0.291	5.014	0.000	Supported

Table AIII.
Structural model

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